
CSS Properties

Basic Font Properties

In CSS, fonts are specified using a set of font-related properties for **typeface**, **size**, **weight**, **font style**, and **special characters**

There are also **shortcut properties** that let you **specify multiple font attributes in a single rule**

Font Properties

`font-family`

`font-size`

`font-weight`

`font-style`

`font-variant`

`font`

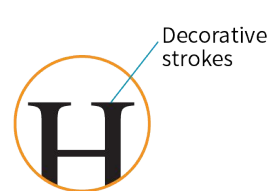
Typeface (Font Family)

font-family

```
p { font-family: "Duru Sans", Verdana, sans-serif; }
```

Generic font families

serif



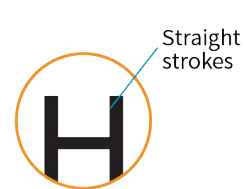
Hello
Times

Hello
Georgia

Hello
Times New Roman

Hello
Lucida

sans-serif



Hello
Verdana

Hello
Trebuchet MS

Hello
Arial

Hello
Arial Black

monospace

W i

Monospace font
(equal widths)

W i

Proportional font
(different widths)

Hello
Courier

Hello
Courier New

Hello
Andale Mono

cursive

Hello
Apple Chancery

Hello
Comic Sans

Hello
Snell

fantasy

Hello
Impact

HELLO
Stencil

HELLO
Mojo

Generic font families

Serif

Examples: Times New Roman, Georgia

Serif typefaces have decorative slab-like appendages (serifs) on the ends of certain letter strokes

Generic font families

sans-serif

Examples: Arial, Verdana, Trebuchet MS, Helvetica

Sans-serif typefaces have straight letter strokes that do not end in serifs

Generic font families

monospace

Examples: `Courier New`

Monospace (also called constant width) **typefaces**, all characters take up the same amount of space on a line

Font stack strategies

The best practice for specifying fonts for web pages is to

start with your first choice,

provide some similar alternatives, and then

end with a generic font family

```
body {  
  font-family: Oswald, Univers, Tahoma, Geneva, sans-serif;  
}
```

Exercise

Create HTML
document inside
`menu.html` file for
the page shown

Black Goose Bistro • Summer Menu

Baker's Corner, Seekonk, Massachusetts

Hours: Monday through Thursday: 11 to 9, Friday and Saturday: 11 to midnight

Appetizers

This season, we explore the spicy flavors of the southwest in our appetizer collection.

Black bean purses

Spicy black bean and a blend of mexican cheeses wrapped in sheets of phyllo and baked until golden. \$3.95

Southwestern napoleons with lump crab — **new item!**

Layers of light lump crab meat, bean and corn salsa, and our handmade flour tortillas. \$7.95

Main courses

Big, bold flavors are the name of the game this summer. Allow us to assist you with finding the perfect wine.

Jerk rotisserie chicken with fried plantains — **new item!**

Tender chicken slow-roasted on the rotisserie, flavored with spicy and fragrant jerk sauce and served with fried plantains and fresh mango. **Very spicy.** \$12.95

Shrimp sate kebabs with peanut sauce

Skewers of shrimp marinated in lemongrass, garlic, and fish sauce then grilled to perfection. Served with spicy peanut sauce and jasmine rice. \$12.95

Grilled skirt steak with mushroom fricasee

Flavorful skirt steak marinated in asian flavors grilled as you like it*. Served over a blend of sauteed wild mushrooms with a side of blue cheese mashed potatoes. \$16.95

* We are required to warn you that undercooked food is a health risk.

Exercise

Make main text of the HTML document you have created appear in **Verdana** or some other **sans-serif** font

```
body {  
    font-family: Verdana, sans-serif;  
}
```

Exercise

Use display font called **Marko One** from Google Web Fonts
(www.google.com/webfonts)

```
<head>
  <meta charset="UTF-8">
  <title>CSS Lab</title>
  <link href="http://fonts.googleapis.com/css?family=Marko+One"
  rel="stylesheet">
  <link href="css/styles.css" rel="stylesheet" >
</head>
```

Exercise

Write a rule that applies Marko One font to the **h1** element

```
h1 {  
    font-family: "Marko One", Georgia, serif;  
}
```

Specifying Font Size

font-size

Values: length unit | percentage | xx-small | x-small | small | medium | large | x-large | xx-large | smaller | larger

Default: medium

The preferred font-size values are **em**, **rem**, and %

Specifying Font Size

This is an example of the default text size in Verdana.

xx-small | x-small | small | medium | large | x-large | **xx-large**

This is an example of the default text size in Times.

xx-small | x-small | small | medium | large | x-large | **xx-large**

Specifying Font Size

Percentage values

To maintain the browser's default size, set the **font-size** of the root element to **100%**

```
html {  
    font-size: 100%;  
}
```


Specifying Font Size

Percentage values

In the following example, the **h1** inherits the default **16px** size from the **html** element, and applying the **150%** value multiplies that inherited value, resulting in an **h1** that is **24 pixels**:

```
h1 {  
    font-size: 150%; /* 150% of 16 = 24 */  
}
```

Exercise

Set the **font-size** of the body element to **100%**

Set the main heading to be **24 pixels** using **em** or **percentage** units

Make the **h2s** the same size as the **body** text

Font Weight (Boldness)

`font-weight`

Values: `normal` | `bold` | `bolder` | `lighter` | `100` | `200` | `300`
| `400` | `500` | `600` | `700` | `800` | `900`

Default: `normal`

An example is shown in the next slide

This is an example of the default text in Verdana.

normal | **bold** | **bolder** | lighter

100 | 200 | 300 | 400 | 500

600 | **700** | **800** | **900**

This is an example of the default text in Times.

normal | **bold** | **bolder** | lighter

100 | 200 | 300 | 400 | 500

600 | **700** | **800** | **900**

Font Style (Italics)

The **font-style** property affects the posture of the text—that is, whether the letter shapes are **vertical (normal)** or **slanted (italic and oblique)**

font-style

Values: `normal` | `italic` | `oblique`

Default: `normal`

Exercise

Make the definition terms `bold` and `italic`

The Shortcut font Property

font

Values: font-style font-weight font-variant font-stretch
font-size/line-height font-family | caption | icon | menu
| message-box | small-caption | status-bar

Default: depends on default value for each property listed

The Shortcut font Property

font

At minimum, the font property must include a font-size value and a font-family value, in that order

```
p {  
  font: 1em sans-serif;  
}
```


The Shortcut font Property

font

At minimum, the font property must include a font-size value and a font-family value, in that order

```
p {  
  font: 1em sans-serif;  
}
```

EXERCISE

Replace all the font properties we've specified for the **h1** element with one declaration with the shorthand font property

```
h1 {  
    font: bold 1.5em "Marko One", Georgia, serif;  
}
```

Changing Text Color

color

Values: color value (name or numeric)

Default: depends on the browser and user's preferences

Changing Text Color

color

```
h1 {  
    color: gray;  
}
```

```
h1 {  
    color: #666666;  
}
```

```
h1 {  
    color: #666;  
}
```

```
h1 {  
    color: rgb(102, 102, 102);  
}
```

Text Line Adjustment

The **line-height** property defines the minimum distance from baseline to baseline in text

line-height

Values: number | length measurement | percentage | normal

Default: normal

Text Line Adjustment

Size of 1em for this text

line-height is set to 2em (twice the text size); the extra space is divided equally above and below the text line, centering it vertically in the line height.

The line-height property defines the minimum distance from baseline to baseline in text.

Baseline A baseline is the imaginary line upon which the bottoms of characters sit. Line height in CSS is similar to leading in traditional typesetting.

```
line-height: 2em;
```

Text Line Adjustment

Example

```
p {  
    line-height: 2;  
}  
  
p {  
    line-height: 2em;  
}  
  
p {  
    line-height: 200%;  
}
```

Indents

text-indent

Values: length measurement | percentage

Default: 0

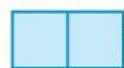
Indents

Example

The effect is shown in the next slide

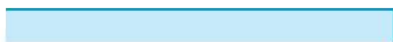
```
p#1 {  
    text-indent: 2em;  
}  
  
p#2 {  
    text-indent: 25%;  
}  
  
p#3 {  
    text-indent: -35px;  
}
```

2em



Paragraph 1. The text-indent property indents only the first line of text by a specified amount. You can specify a length measurement or a percentage value.

25%



Paragraph 2. The text-indent property indents only the first line of text by a specified amount. You can specify a length measurement or a percentage value.

-35px



Paragraph 3. The text-indent property indents only the first line of text by a specified amount. You can specify a length measurement or a percentage value.

Horizontal Text Alignment

text-align

Values: left | right | center | justify | start | end

Default: start

An example is shown in the next slide

`text-align: left;`

Paragraph 1. The text-align property controls the horizontal alignment of the text within an element. It does not affect the alignment of the element on the page. The resulting text behavior of the various values should be fairly intuitive.

`text-align: right;`

Paragraph 2. The text-align property controls the horizontal alignment of the text within an element. It does not affect the alignment of the element on the page. The resulting text behavior of the various values should be fairly intuitive.

`text-align: center;`

Paragraph 3. The text-align property controls the horizontal alignment of the text within an element. It does not affect the alignment of the element on the page. The resulting text behavior of the various values should be fairly intuitive.

`text-align: justify;`

Paragraph 4. The text-align property controls the horizontal alignment of the text within an element. It does not affect the alignment of the element on the page. The resulting text behavior of the various values should be fairly intuitive.

Text Decoration

text-decoration

Values: none | underline | overline | line-through | blink

Default: none

Text Decoration

Examples

I've got laser eyes.

```
text-decoration: underline;
```

I've got laser eyes.

```
text-decoration: overline;
```

~~I've got laser eyes.~~

```
text-decoration: line-through;
```

Text Capitalization

text-transform

Values: none | capitalize | lowercase | uppercase |
full-width

Default: none

Check next slide or example

Text Capitalization

Examples

text-transform: none;
(as it was typed in the source)

And I know what you're thinking.

text-transform: capitalize;

And I Know What You're Thinking.

text-transform: lowercase;

and i know what you're thinking.

text-transform: uppercase;

AND I KNOW WHAT YOU'RE THINKING.

Letter Spacing

letter-spacing

Values: length measurement | normal

Default: normal

Letter Spacing

Examples

B l a c k G o o s e B i s t r o S u m m e r M e n u

```
p { letter-spacing: 8px; }
```

Black Goose Bistro Summer Menu

```
p { word-spacing: 1.5em; }
```

List Markers

list-style-type

Values: none | disc | circle | square | decimal |
decimal-leading-zero | lower-alpha | upper-alpha |
lower-latin | upper-latin | lower-roman | upper-roman |
lower-greek

Default: disc

Numbering and Lettering Styles

Keyword	System
<code>decimal</code>	1, 2, 3, 4, 5, ...
<code>decimal-leading-zero</code>	01, 02, 03, 04, 05, ...
<code>lower-alpha</code> or <code>lower-latin</code>	a, b, c, d, ...
<code>upper-alpha</code> or <code>upper-latin</code>	A, B, C, D, ...
<code>lower-roman</code>	i, ii, iii, iv, v, ...
<code>upper-roman</code>	I, II, III, IV, V, ...
<code>lower-greek</code>	α , β , γ , δ , ε , ...

List Item Display Role

Any element can perform like a list item by setting its **display** property to **list-item**

```
p.lettered {  
    display: list-item;  
    list-style-type:  
upper-alpha;  
}
```

List Marker Position

list-style-position

Values: inside | outside | hanging

Default: outside

List Marker Position

outside

- **Radish.** Praesent in lacinia risus. Morbi urna ipsum, efficitur id erat pellentesque, tincidunt commodo sem. Phasellus est velit, porttitor vel dignissim vitae, commodo ut urna.
- **Avocado.** Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Curabitur lacinia accumsan est, ut malesuada lorem consectetur eu.
- **Pomegranite.** Nam euismod a ligula ac bibendum. Aenean ac justo eget lorem dapibus aliquet. Vestibulum vitae luctus orci, id tincidunt nunc. In a mauris odio. Duis convallis enim nunc.

inside

- **Radish.** Praesent in lacinia risus. Morbi urna ipsum, efficitur id erat pellentesque, tincidunt commodo sem. Phasellus est velit, porttitor vel dignissim vitae, commodo ut urna.
- **Avocado.** Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Curabitur lacinia accumsan est, ut malesuada lorem consectetur eu.
- **Pomegranite.** Nam euismod a ligula ac bibendum. Aenean ac justo eget lorem dapibus aliquet. Vestibulum vitae luctus orci, id tincidunt nunc. In a mauris odio. Duis convallis enim nunc.

Custom Bullets

list-style-image

Values: `url(location) | none`

Default: `none`

Custom Bullet

Example

```
ul {  
  list-style-type: disc;  
  list-style-image: url(/images/custom_bullet.gif);  
  list-style-position: outside;  
}
```

Foreground Color

The **foreground** of an element consists of its text and border (if one is specified)

You specify a foreground color with the **color** property

Background Color

background-color

Values: color value (name or numeric) | transparent

Default: transparent

Background Color

A background color **fills the canvas behind the element** that includes the **content area**, and any **padding** (extra space) added around the content, extending **behind the border** out to its outer edge

Background will **not** extend into the margin

Note: background colors do not inherit, but because the default **background setting for all elements is transparent**, the parent's background color shows through its descendant elements

Example of adding background color is shown in the next slide

Background Color

```
blockquote {  
  border: 4px dashed;  
  color: green;  
  background-color: #c6de89;  
}
```

In the latitude of central New England, cabbages are not secure from injury from frost with less than a foot of earth thrown over the heads. In mild winters a covering of half that depth will be sufficient; but as we have no prophets to foretell our mild winters, a foot of earth is safer than six inches.

Background Clip

background-clip

Values: border-box | padding-box | content-box

Default: border-box

An example is shown in the next slide

```
blockquote {  
  padding: 1em; border: 4px dashed; color: green; background-color: #C6DE89;}  
}
```

In the latitude of central New England,
cabbages are not secure from injury from
frost with less than a foot of earth thrown
over the heads.

background-clip: `border-box`;

In the latitude of central New England,
cabbages are not secure from injury from
frost with less than a foot of earth thrown
over the heads.

background-clip: `padding-box`;

In the latitude of central New England,
cabbages are not secure from injury from
frost with less than a foot of earth thrown
over the heads.

background-clip: `content-box`;

Opacity

Opacity is another way of adding level of transparency to an element

The opacity setting applies to the entire element — both the **foreground** and the **background**

opacity

Values: number (0 to 1)

Default: 1

An example is shown in the next slide


```
h1 {  
  color: gold;  
  background: white;  
  opacity: 0.25;  
}  
h1 {  
  color: gold;  
  background: white;  
  opacity: 0.5;  
}  
h1 {  
  color: gold;  
  background: white;  
  opacity: 1;  
}
```

Playing with opacity

opacity: .25;

Playing with opacity

opacity: .5;

Playing with opacity

opacity: 1;

Exercises

Make the **h1** heading purple (R:153, G:51, B:153, or #993399)

Make the **background** of the entire page a light green (R:210, G:220, B:157, or #d2dc9d)

Make the **background** of the header white with 50% transparency (R:255, G:255, B:255, .5)

Pseudo Class Selectors

Pseudo-class selectors are indicated by the colon (:) character

They typically go immediately after an element name as follows

li:first-child

Link Pseudo-Classes

:link

Applies a style to unclicked (unvisited) links

:visited

Applies a style to links that have already been clicked

By default, browsers typically display linked text as **blue** and links that have been clicked as **purple**

User Action Pseudo-Classes

- :focus** Applies when the element is selected and ready for input
- :hover** Applies when the mouse pointer is over the element
- :active** Applies when the element (such as a link or button) is in the process of being clicked or tapped

Applying All Pseudo-Classes

When you apply styles to a elements with all five pseudo-classes, the order in which they appear is important for them to function properly

The required order for pseudo-classes is:

:link

:visited

:focus

:hover

:active

Pseudo-Classes Example

```
a { text-decoration: none; }  
a:link { color: maroon; }  
a:visited { color: gray; }  
a:focus { color: maroon; background-color: #ffd9d9; }  
a:hover { color: maroon; background-color: #ffd9d9; }  
a:active { color: red; background-color: #ffd9d9; }
```

Pseudo-Classes

Example

Samples of my work:

- Pen and Ink Illustrations
- Paintings
- Collage

`a:link`

Links are maroon and not underlined.

Samples of my work:

- Pen and Ink Illustrations
- Paintings
- Collage

`a:focus`

`a:hover`

While the mouse is over the link or when the link has focus, the pink background color appears.

Samples of my work:

- Pen and Ink Illustrations
- Paintings
- Collage

`a:active`

As the mouse button is being pressed, the link turns bright red.

Samples of my work:

- Pen and Ink Illustrations
- Paintings
- Collage

`a:visited`

After that link has been visited, the link is gray.

Exercises

Write a rule that makes **links** purple (#939)

When the mouse is placed over **links**, make the text a brighter purple (#c700f2) and add a white background color (#fff)

Use these same style rules for when the **links** are in focus

As the mouse is being clicked (or tapped on a touch device), add a white background color and make the text turn a vibrant purple (#ff00ff)

Other CSS3 Pseudo-Classes

```
:root , :empty, :first-child, :last-child, :only-child,  
:first-of-type, :last-of-type, :only-of-type, :nth-child(),  
:nth-last-child(), :nth-of-type(), :nth-last-of-type(),  
:enabled, :disabled, :checked, :target (fragment  
identifier), :lang(), :not()
```

Pseudo Element Selectors

Pseudo-elements are indicated by a double colon (: :)

Pseudo Element Selectors

`::first-line`

This selector applies style rule for the following properties to the **first line** of the specified element

color, font, background, word-spacing, letter-spacing,
text-decoration, vertical-align, text-transform,
line-height

Pseudo Element Selectors

::first-letter

This selector applies style rule for the following properties to the **first letter** of the specified element

color, font, background, word-spacing, letter-spacing,
text-decoration, vertical-align(if float is none),
text-transform, padding, margin, border, line-height,
float

First Line and First Letter Example

```
p::first-line {  
  letter-spacing: 9px;  
}  
  
p::first-letter {  
  font-size: 300%;  
  color: orange;  
}
```

First Line and First Letter Example

::first-line I n s o m e o f t h e b e s t c a b b a g e -
growing sections of the country, until within a comparatively few
years it was the very general belief that cabbage would not do well
on upland. Accordingly the cabbage patch would be found on the
lowest tillage land of the farm.

::first-letter **I**n some of the best cabbage-growing sections of the country, until
within a comparatively few years it was the very general belief that
cabbage would not do well on upland. Accordingly the cabbage
patch would be found on the lowest tillage land of the farm.

Pseudo Element Selectors

Generated Content with `::before` and `::after`

Generated contents are contents that browsers insert on the fly

Using the `::before` and `::after` pseudo-elements you can tell browsers to generate content before or after any element

Pseudo Element Selectors

The following example shows how to insert an image by using the `url()` function before the paragraph and `"Thank you."` at the end of the paragraph

```
p.warning::before {  
  content: url(exclamation.png);  
  margin-right: 6px;  
}
```

```
p.warning::after {  
  content: " Thank you.";  
  color: red;  
}
```

Pseudo Element Selectors

The following example shows how to insert an image by using the `url()` function before the paragraph and `"Thank you."` at the end of the paragraph

```
<p class="warning">We are required to warn you that  
undercooked food is a health risk.</p>
```



We are required to warn you that undercooked food is a health risk. **Thank you.**

Attribute Selectors

Attribute selectors **target elements based on attribute names or values**

Attribute Selectors

Simple Attribute Selector targets elements with a particular attribute

`element[attribute]`

```
img[title] {border: 3px solid;}
```

Attribute Selectors

Exact Value Selector selects elements with a specific value for the attribute

`element[attribute="exact value"]`

```
img[title="first grade"] {border: 3px solid;}
```

Attribute Selectors

Partial Attribute Value Selector (indicated with a bar, ~) selects elements by partial attribute value

element[attribute~="value"]

```
img[title~="grade"] {border: 3px solid;}
```

Attribute Selectors

Hyphen Separated Attribute Value Selector (indicated with a bar, |) targets hyphen-separated values

`element[attribute]="value"]`

```
[hreflang]="en"] {border: 3px solid;}
```

Attribute Selectors

Beginning Substring Attribute Value Selector (indicated with a carat, ^) matches elements whose specified attribute values start in the string of characters in the selector

`element[attribute^="first part of the value"]`

```
img[src^="/images/icons"] {border: 3px solid;}
```


Attribute Selectors

Ending Substring Attribute Value Selector (indicated with a dollar sign, \$) matches elements whose specified attribute values end in the string of characters in the selector

`element[attribute$="last part of the value"]`

```
a[href$=".pdf"] {border-bottom: 3px solid;}
```

Attribute Selectors

Arbitrary Substring Attribute Value Selector (indicated with an asterisk, *) looks for the provided text string in any part of the attribute value specified

`element[attribute*="any part of the value"]`

```
img[title*="February"] {border: 3px solid;}
```

Background Properties

background-color

background-image

background-repeat

background-position

background-attachment

background-clip

background-size

background

Background Image

Most decorative images are added to pages and elements as backgrounds with CSS

Background Image

background-image

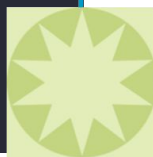
Values: `url(location of image) | none`

Default: `none`

Background Image

```
body {  
  background-image: url(star.png)  
}
```

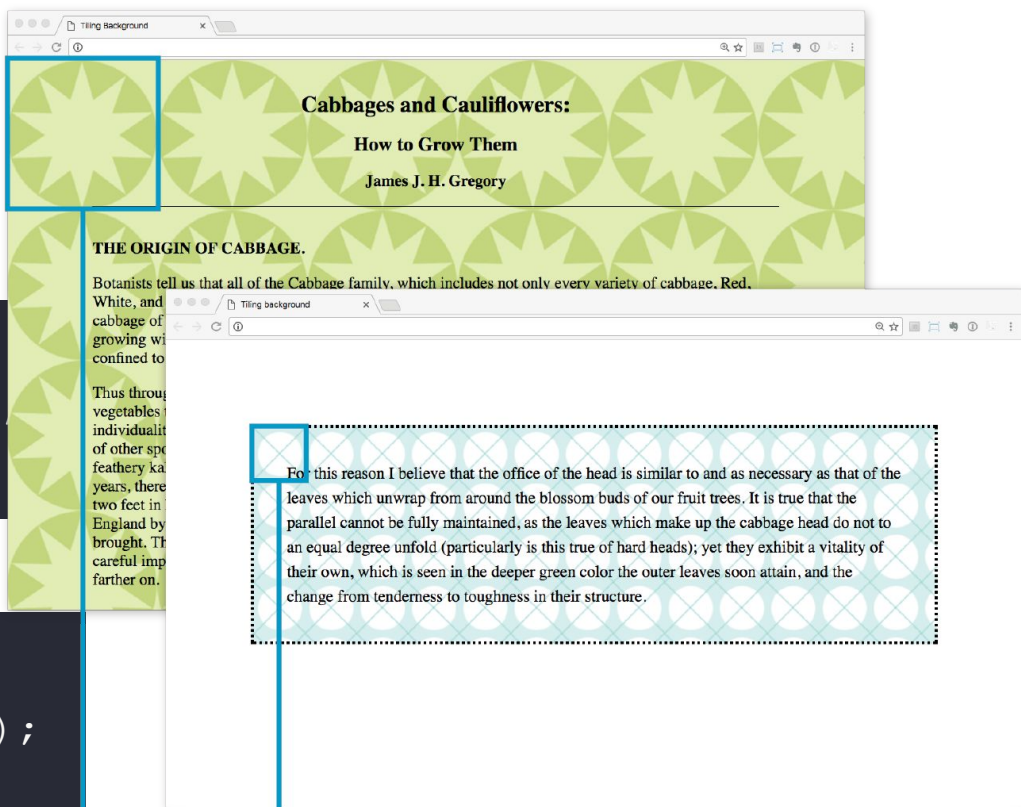
```
blockquote {  
  background-image: url(dot.png);  
  padding: 2em;  
  border: 4px dashed;  
}
```



star.png
150 × 150 pixels



dot.png
50 × 50 pixels



Exercise

Add background image to your page

Background Repeat

background-repeat

Values: `repeat` | `no-repeat` | `repeat-x` | `repeat-y` | `space` |
`round`

Default: `repeat`

Background Repeat

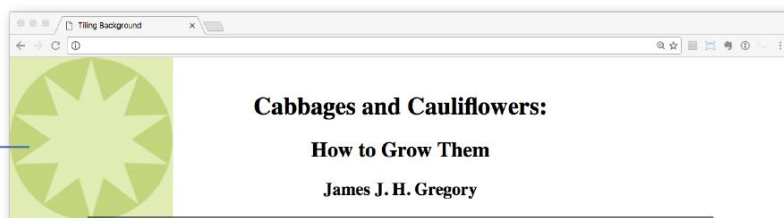
```
body {  
  background-image: url(star.png);  
  background-repeat: repeat-y;  
}
```

```
body {  
  background-image: url(star.png);  
  background-repeat: no-repeat;  
}
```

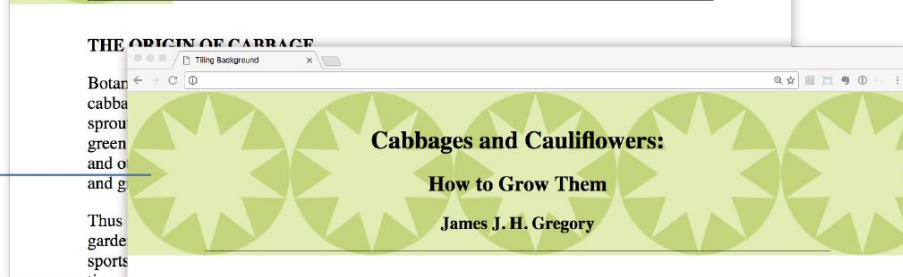
```
body {  
  background-image: url(star.png);  
  background-repeat: repeat-x;  
}
```

Background Repeat

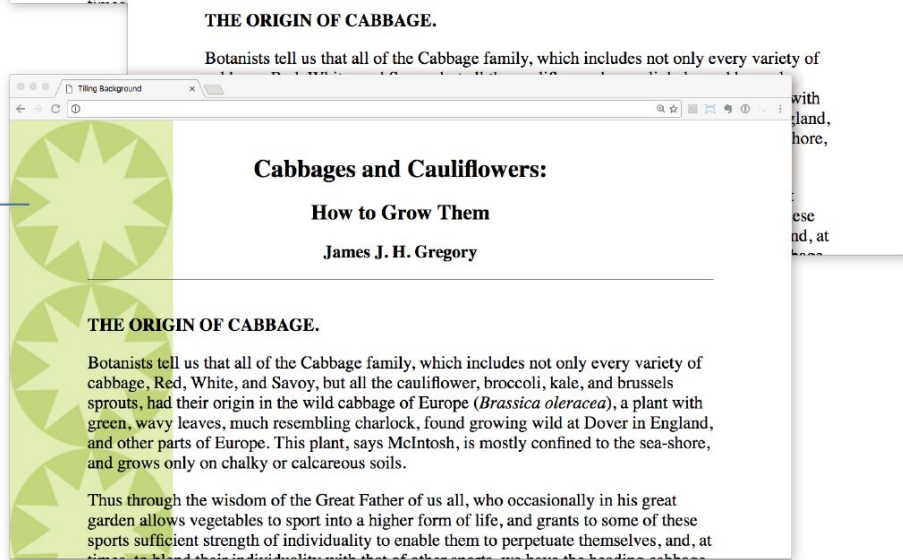
no-repeat



repeat-x



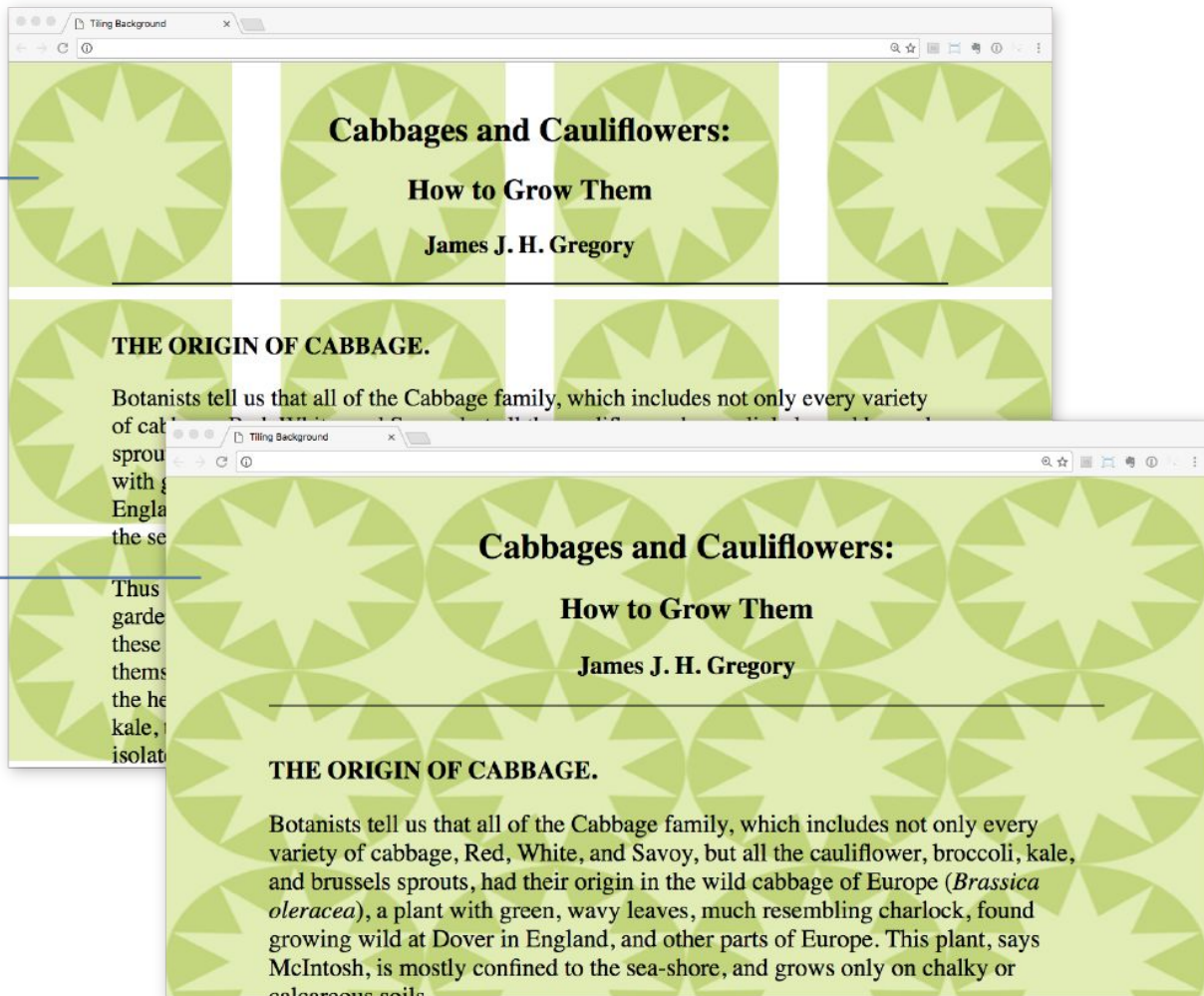
repeat-y



space and round

space

round



Exercise

Experiment **background-repeat** property

Example

```
header {  
  margin-top: 0;  
  padding: 3em 1em 2em 1em;  
  text-align: center;  
  background-color: rgba(255, 255, 255, 0.5);  
  background-image: url(images/purpledots.png);  
  background-repeat: repeat-x;  
}
```

Background Position

background-position

Values: length measurement | percentage | left | center |
right | top | bottom

Default: 0% 0% (same as left top)

Background Position

For this reason I believe that the office of the head is similar to and as necessary as that of the leaves which unwrap from around the blossom buds of our fruit trees. It is true that the parallel cannot be fully maintained, as the leaves which make up the cabbage head do not to an equal degree unfold (particularly is this true of hard heads); yet they exhibit a vitality of their own, which is seen in the deeper green color the outer leaves soon attain, and the change from tenderness to toughness in their structure.

background-position: `left bottom`;

For this reason I believe that the office of the head is similar to and as necessary as that of the leaves which unwrap from around the blossom buds of our fruit trees. It is true that the parallel cannot be fully maintained, as the leaves which make up the cabbage head do not to an equal degree unfold (particularly is this true of hard heads); yet they exhibit a vitality of their own, which is seen in the deeper green color the outer leaves soon attain, and the change from tenderness to toughness in their structure.

background-position: `right center`;

or

background-position: `right`;

300px
50px

For this reason I believe that the office of the head is similar to and as necessary as that of the leaves which unwrap from around the blossom buds of our fruit trees. It is true that the parallel cannot be fully maintained, as the leaves which make up the cabbage head do not to an equal degree unfold (particularly is this true of hard heads); yet they exhibit a vitality of their own, which is seen in the deeper green color the outer leaves soon attain, and the change from tenderness to toughness in their structure.

background-position: `300px 50px`;

25%

For this reason I believe that the office of the head is similar to and as necessary as that of the leaves which unwrap from around the blossom buds of our fruit trees. It is true that the parallel cannot be fully maintained, as the leaves which make up the cabbage head do not to an equal degree unfold (particularly is this true of hard heads); yet they exhibit a vitality of their own, which is seen in the deeper green color the outer leaves soon attain, and the change from tenderness to toughness in their structure.

25%

background-position: `25% 100%`;

Background Position Origin

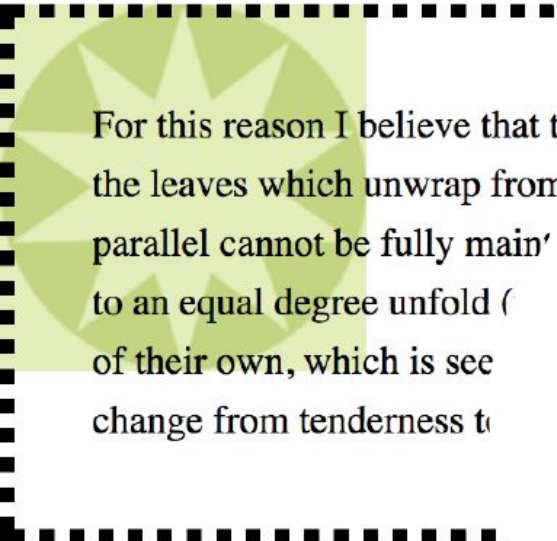
background-origin

Values: `border-box` | `padding-box` | `content-box`

Default: `padding-box`

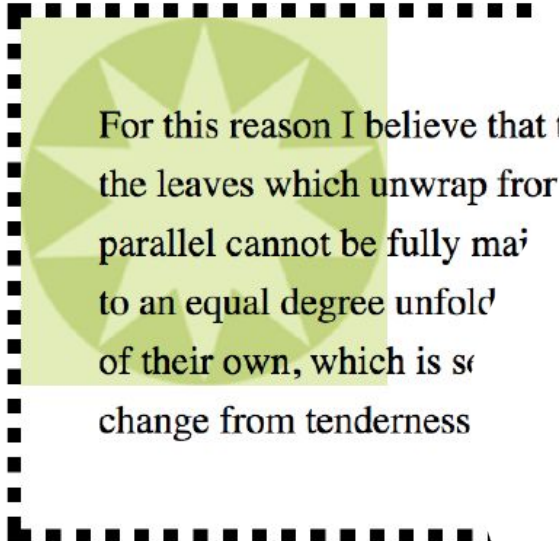
Background Position Origin

border-box

A diagram illustrating the border-box model. It features a light green square with a white starburst pattern in the center. This square is enclosed within a thick, black dashed border. The text is positioned inside the green square, centered horizontally and vertically.

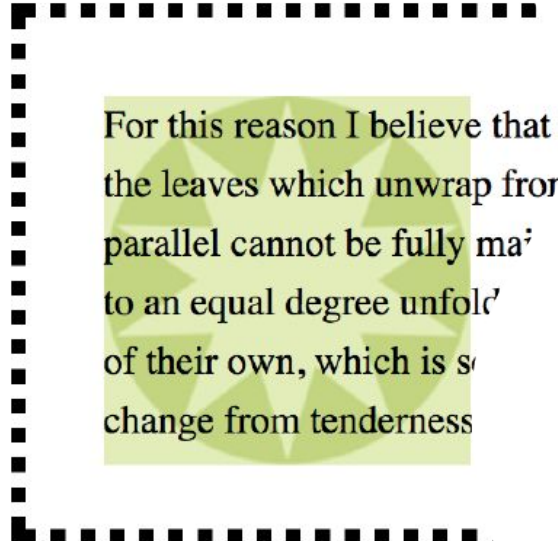
For this reason I believe that the
the leaves which unwrap from
parallel cannot be fully main'
to an equal degree unfold ('
of their own, which is see
change from tenderness to

padding-box

A diagram illustrating the padding-box model. It features a light green square with a white starburst pattern in the center. This square is enclosed within a thick, black dashed border. The text is positioned inside the green square, centered horizontally and vertically.

For this reason I believe that t
the leaves which unwrap from
parallel cannot be fully ma'
to an equal degree unfold'
of their own, which is s
change from tenderness

content-box

A diagram illustrating the content-box model. It features a light green square with a white starburst pattern in the center. This square is enclosed within a thick, black dashed border. The text is positioned inside the green square, centered horizontally and vertically.

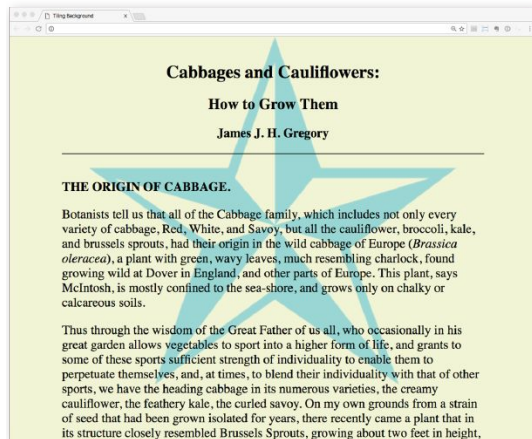
For this reason I believe that t
the leaves which unwrap from
parallel cannot be fully ma'
to an equal degree unfold'
of their own, which is s
change from tenderness

Background Attachment

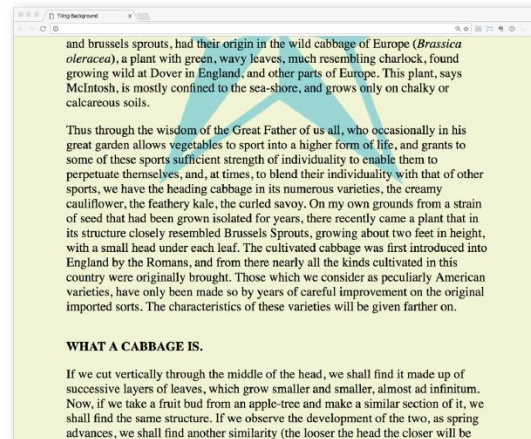
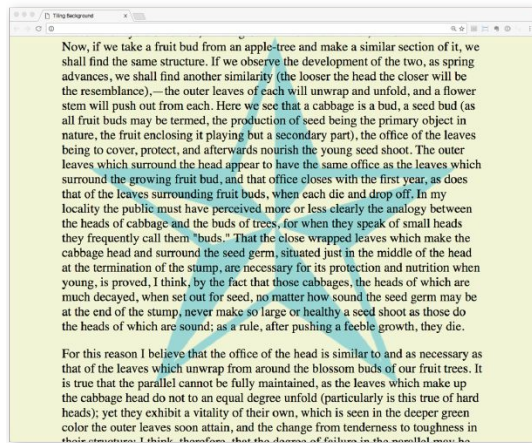
background-attachment

Values: `scroll` | `fixed` | `local`

Default: `scroll`



A large non-repeating background image in the body of the document.



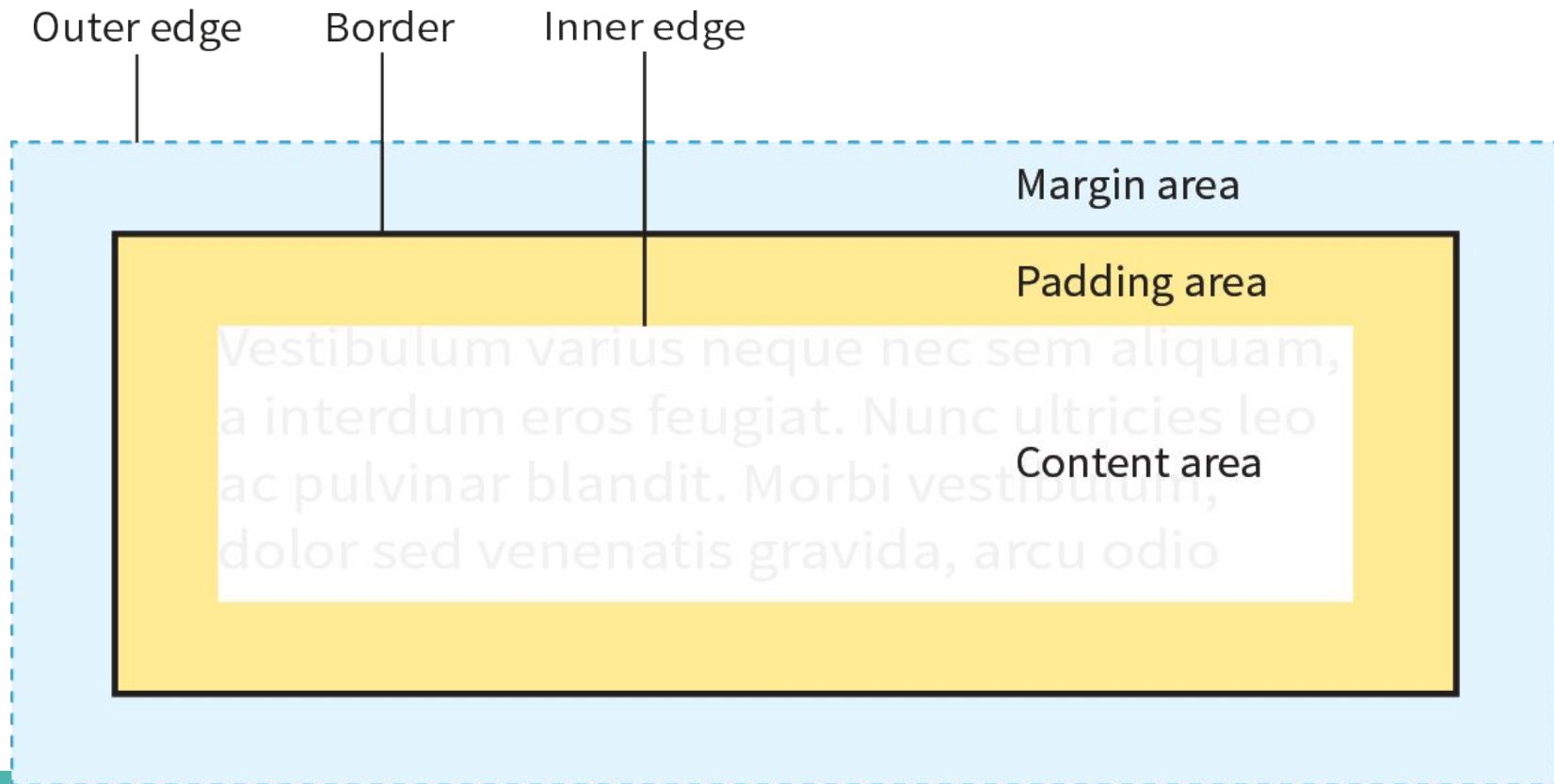
background-attachment: **scroll**;

By default, the background image is attached to the body element and scrolls off the page when the page content scrolls.

background-attachment: **fixed**;

When background-attachment is set to fixed, the image stays in its position relative to the browser viewing area and does not scroll with the content.

The Box Model



Specifying Box Dimension

width

Values: `length` | `percentage` | `auto`

Default: `auto`

height

Values: `length` | `percentage` | `auto`

Default: `auto`

Specifying Box Dimension

You can specify the width and height only for block-level elements and non-text inline elements such as images

Specifying Box Dimension

There are two ways to specify the size of an element

The default method or by setting **content-box** value to **box-sizing** property - applies the width and height values to the content box

By setting **border-box** value to **box-sizing** - applies the width and height values to the border box, which includes the content, padding, and border

Specifying Box Dimension

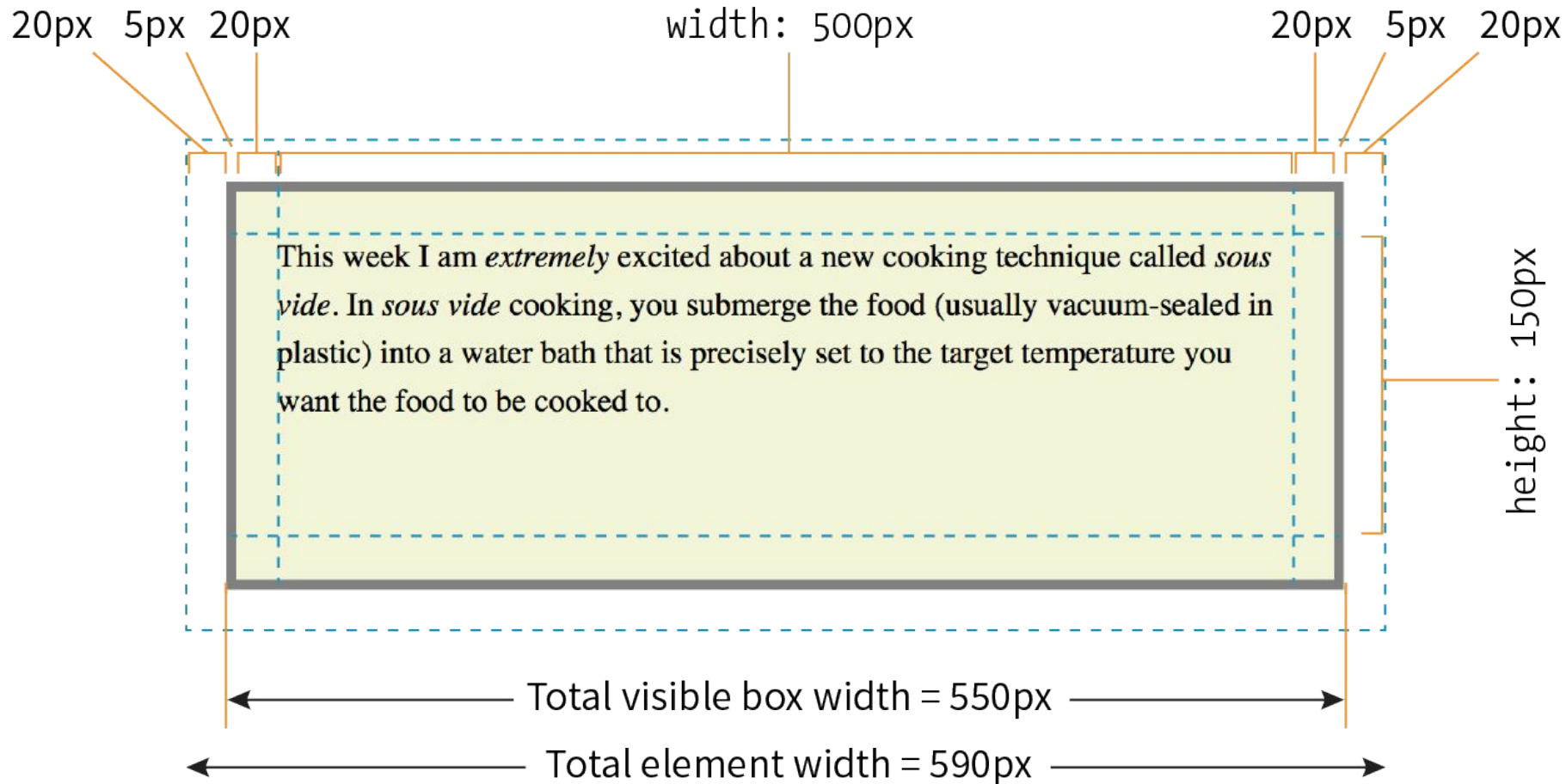
Using the Content Box

```
p {  
  background: #f2f5d5;  
  width: 500px;  
  height: 150px;  
  padding: 20px;  
  border: 5px solid gray;  
  margin: 20px;  
}
```

Exercise

Using **Chrome DevTools** calculate the visible and total width and height values of the element

content-box model

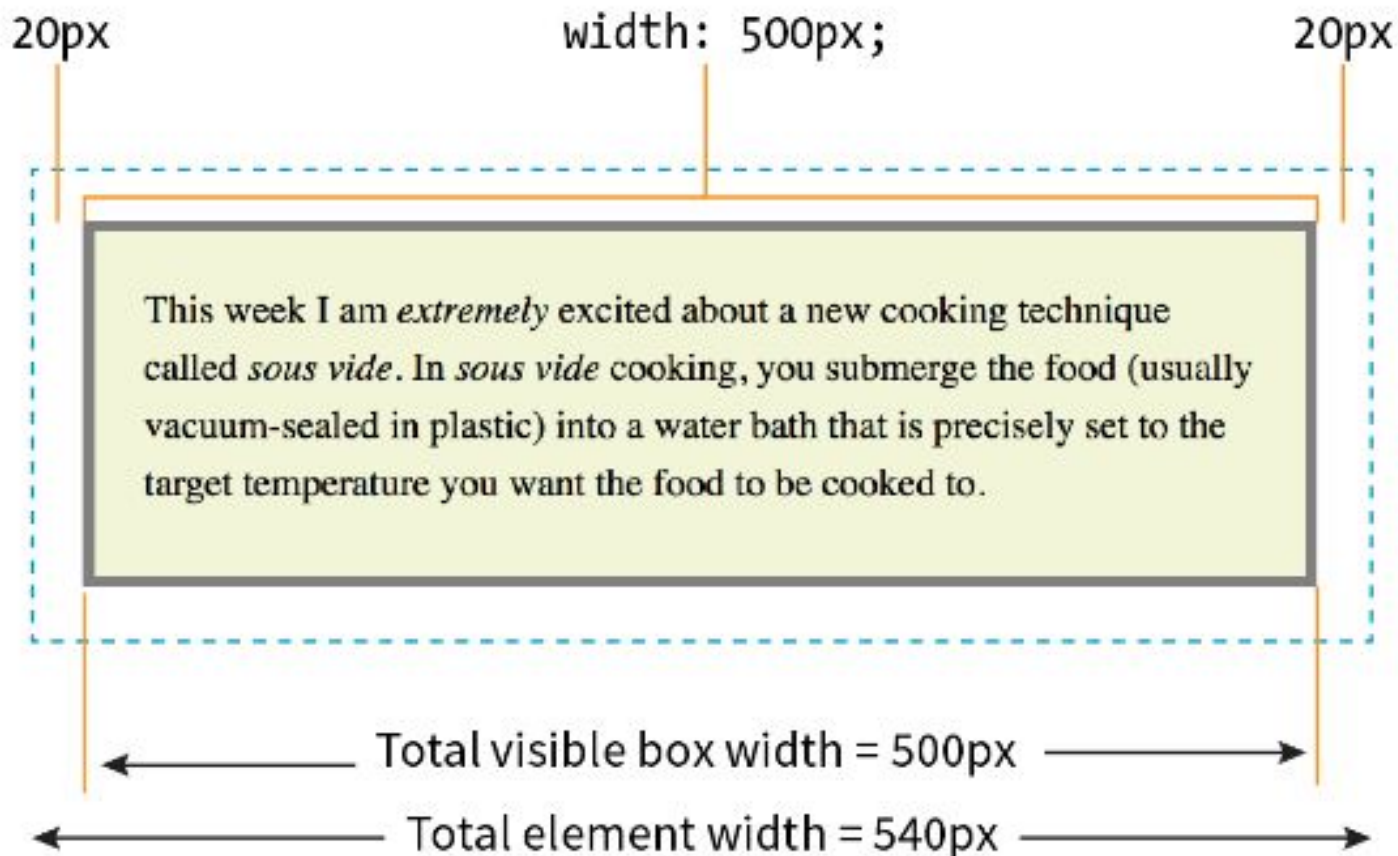


Specifying Box Dimension

Using the Border Box

```
p {  
  background: #f2f5d5;  
  width: 500px;  
  height: 150px;  
  padding: 20px;  
  border: 5px solid gray;  
  margin: 20px;  
  box-sizing: border-box;  
}
```

border-box model



Exercise

Using **Chrome DevTools** calculate the visible and total width and height values of the element

```
box-sizing: content-box;  
width: 500px;
```

This week I am *extremely* excited about a new cooking technique called *sous vide*. In *sous vide* cooking, you submerge the food (usually vacuum-sealed in plastic) into a water bath that is precisely set to the target temperature you vacuum-sealed in plastic) into a water bath that is precisely set to the target temperature you want the food to be cooked to.

```
box-sizing: border-box;  
width: 500px;
```

Handling Overflow

When an element is sized too small for its contents, you can specify what to do with the content that doesn't fit by using the **overflow** property

overflow

Values: `visible` | `hidden` | `scroll` | `auto`

Default: `visible`

visible

Applying the masks to the glasses is the most labor-intensive part of the process. Not only do you have to measure, place, and burnish on each mask, but you also need to completely cover the remainder of the glass in heavy paper. Any exposed areas (even inside) will get scratched by the flying sand, so it has to be a good seal.

hidden

Applying the masks to the glasses is the most labor-intensive part of the process. Not only do you have to measure, place, and burnish on each mask, but you also need to completely cover the remainder of the glass

auto (short text)

Applying the masks to the glasses is the most labor-intensive part of the process.

scroll

labor-intensive part of the process. Not only do you have to measure, place, and burnish on each mask, but you also need to completely cover the remainder of the glass in heavy paper. Any exposed areas (even

auto (long text)

Applying the masks to the glasses is the most labor-intensive part of the process. Not only do you have to measure, place, and burnish on each mask, but you also need to completely cover the remainder of the glass

Padding

Padding is the space between the content area and the border

padding

Values: `length` | `percentage`

Default: `0`

Padding

Padding is the space between the content area and the border

padding-top, padding-right, padding-bottom, padding-left

Values: `length` | `percentage`

Default: 0

Padding

```
blockquote {  
  padding-top: 2em;  
  padding-right: 4em;  
  padding-bottom: 2em;  
  padding-left: 4em;  
  background-color: #d098d4;  
}
```

Padding

4em 2em
This week I am *extremely* excited about a new 4em
cooking technique called *sous vide*. In *sous vide*
cooking, you submerge the food (usually vacuum-
sealed in plastic) into a water bath that is
precisely set to the target temperature you want
the food to be cooked to. 2em

The Shorthand padding Property

`padding: top right bottom left; // clockwise direction`

```
blockquote {  
  padding: 2em 4em 2em 4em;  
  background-color: #d098d4;  
}
```

The Shorthand padding Property

If the left and right padding are the same, you can shorten it by supplying only three values

padding: top right/left bottom;

```
blockquote {  
  padding: 2em 4em 2em;  
  background-color: #d098d4;  
}
```

The Shorthand padding Property

If the top and bottom padding are the same and right and left padding are the same, you can shorten it by supplying only two values

padding: top/bottom right/left;

```
blockquote {  
  padding: 2em 4em;  
  background-color: #d098d4;  
}
```

The Shorthand padding Property

If all sides padding are the same, you can shorten it by supplying only one values

padding: top/bottom/right/left;

```
blockquote {  
  padding: 2em;  
  background-color: #d098d4;  
}
```

Exercise

Create the page shown

Use HTML5's semantic elements

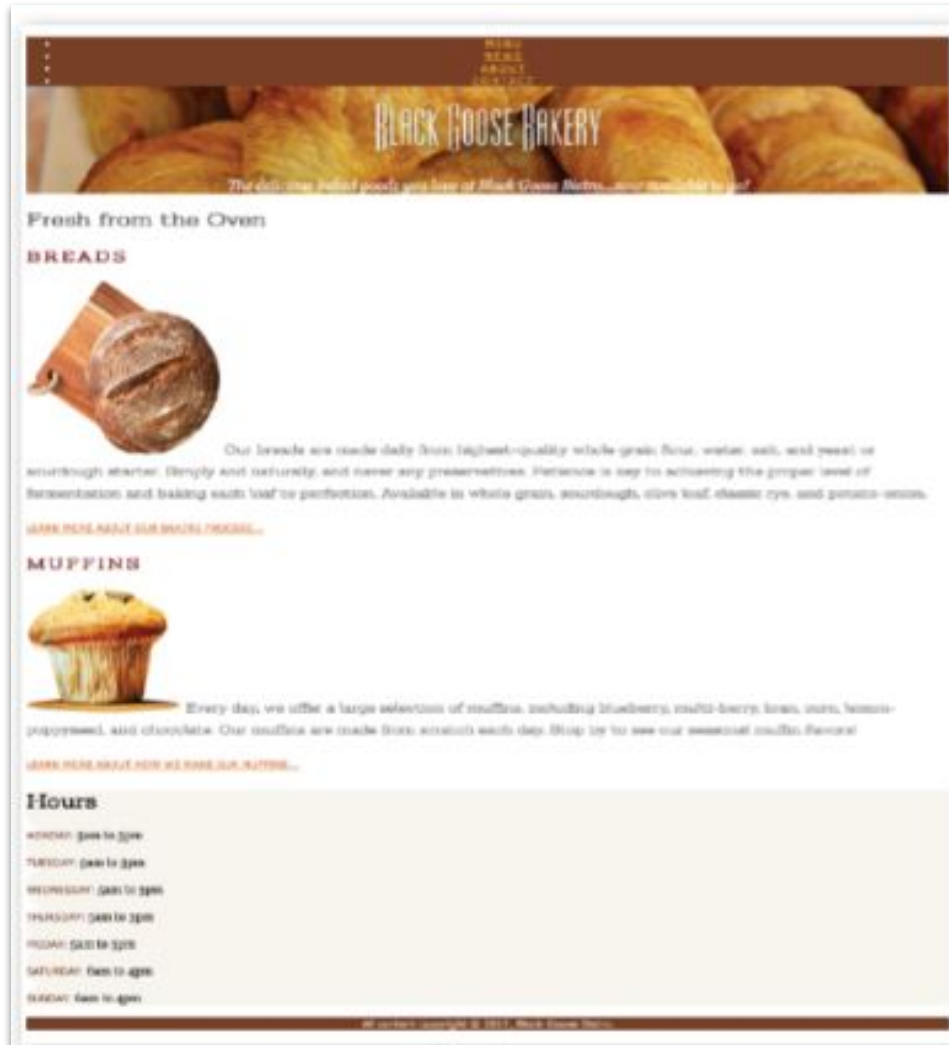
<header>

<main>

<section>

<aside>

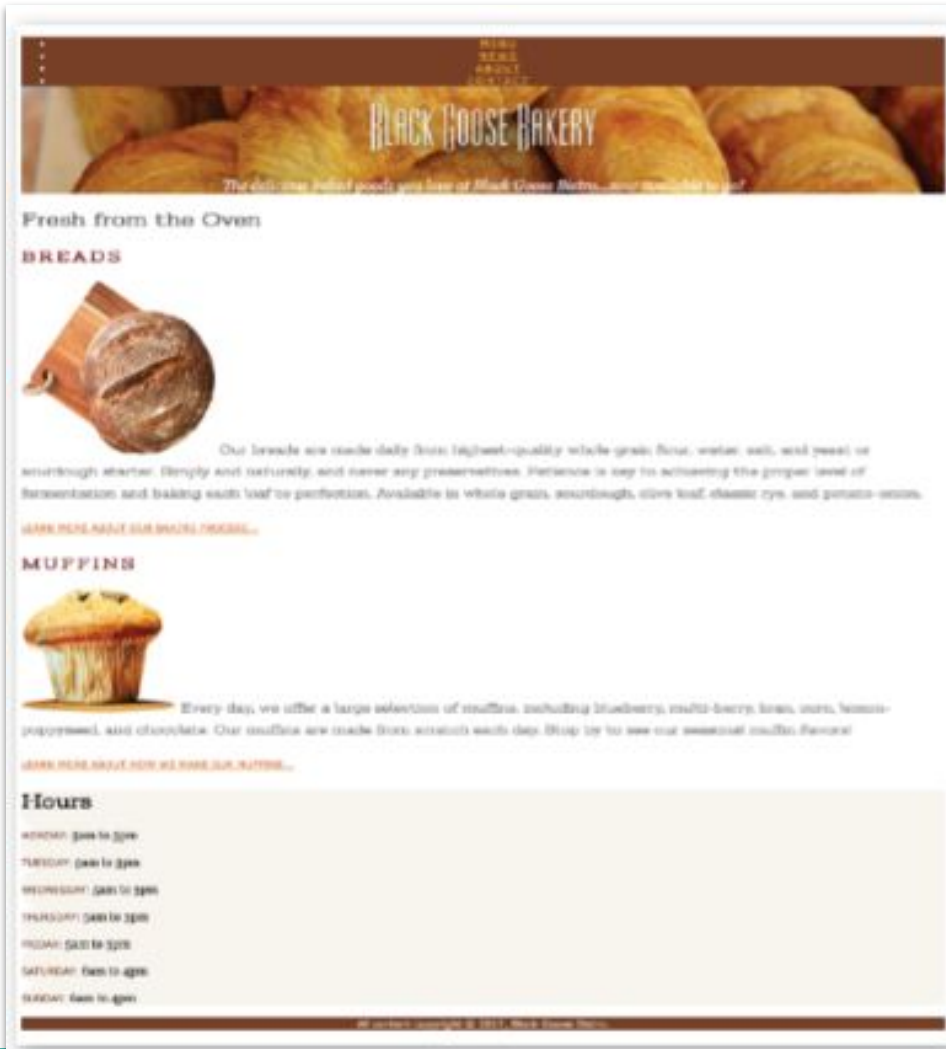
<footer>



Exercise

Set the **box-sizing** model to **border-box** for all the elements in the document

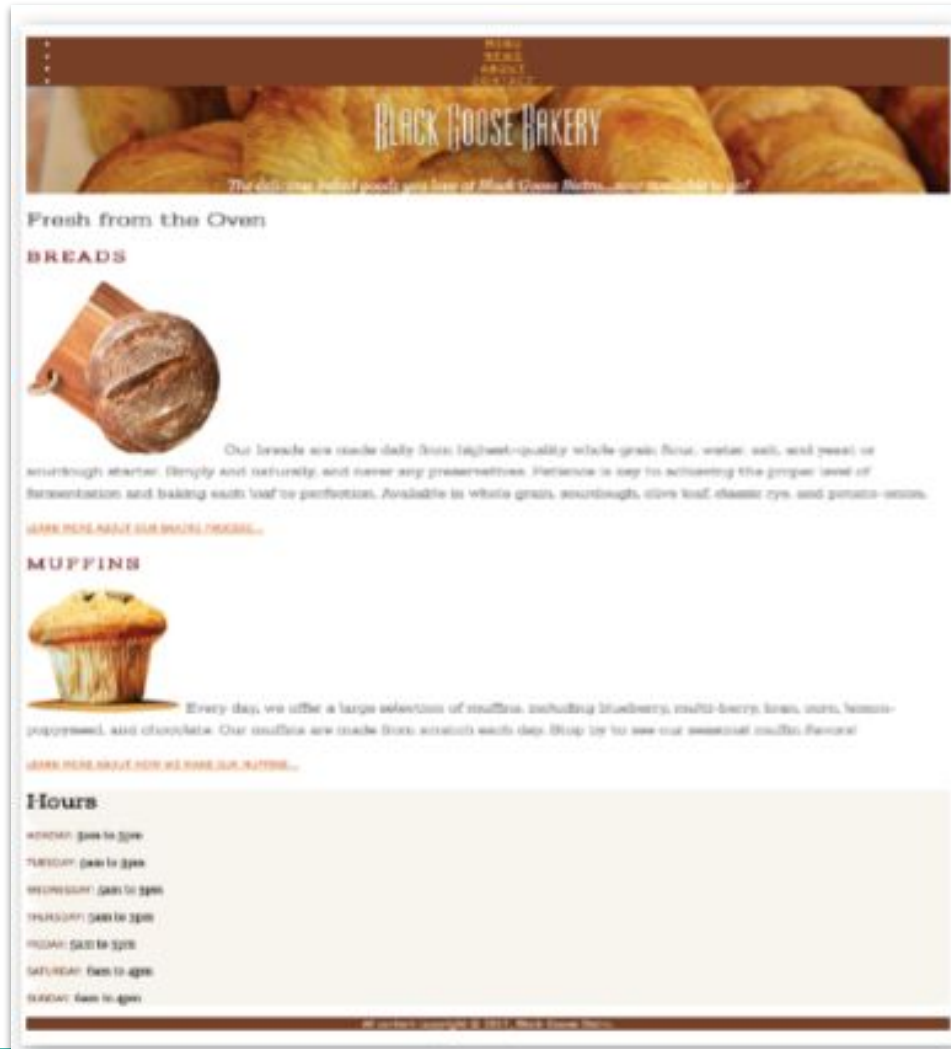
```
html {  
  box-sizing: border-box;  
}  
  
* {  
  box-sizing: inherit;  
}
```



Exercise

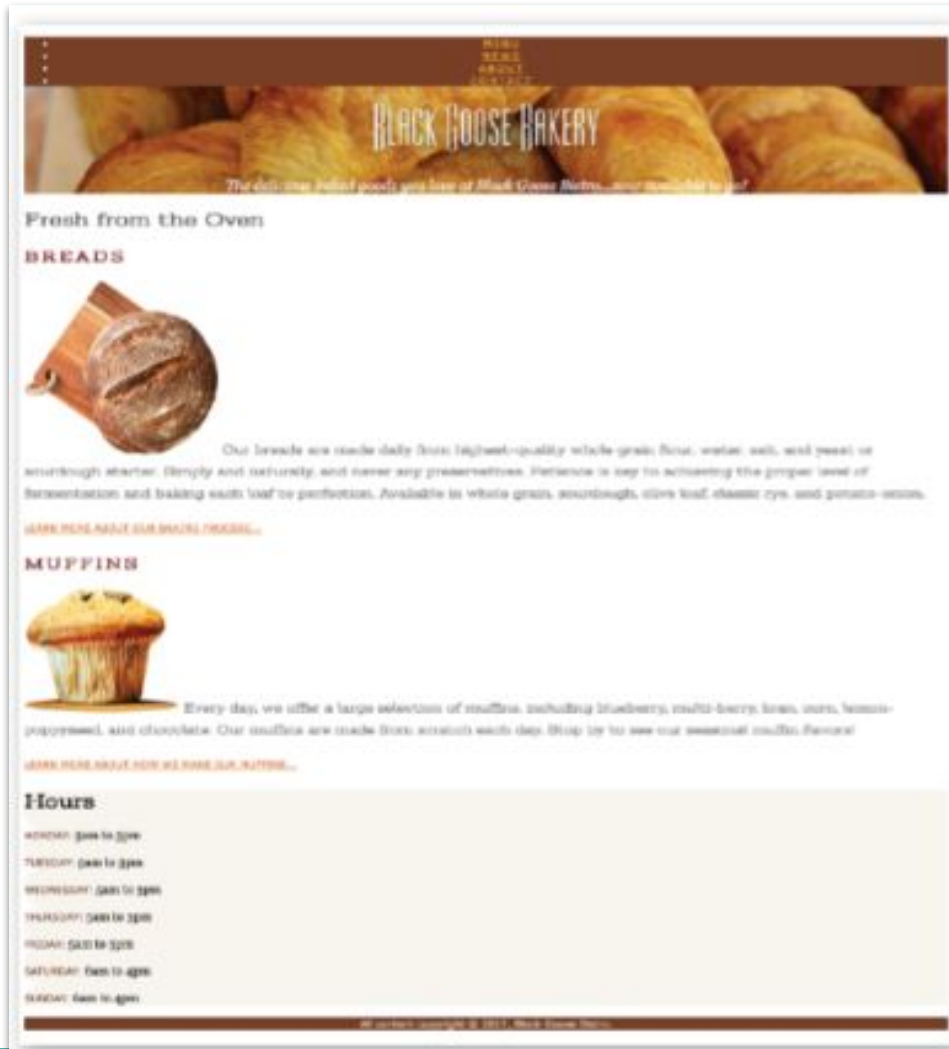
Add **1em** of padding on all sides of **main** element

```
main {  
  padding: 1em;  
}
```



Exercise

Add **1em** of padding on all sides of **aside** element except the left side and add **45 pixels** of padding on the left side



Borders

A border is a line drawn around the content area and its (optional) padding

border-style

Values: none | solid | hidden | dotted | dashed | double |
groove | ridge | inset | outset

Default: none

Borders

A border is a line drawn around the content area and its (optional) padding

**`border-top-style`, `border-right-style`, `border-bottom-style`,
`border-left-style`**

Values: `none` | `solid` | `hidden` | `dotted` | `dashed` | `double` |
`groove` | `ridge` | `inset` | `outset`

Default: `none`

none

solid

double

dotted

dashed

groove

ridge

inset

outset

Borders

If you do not specify a width, the default medium width will be used

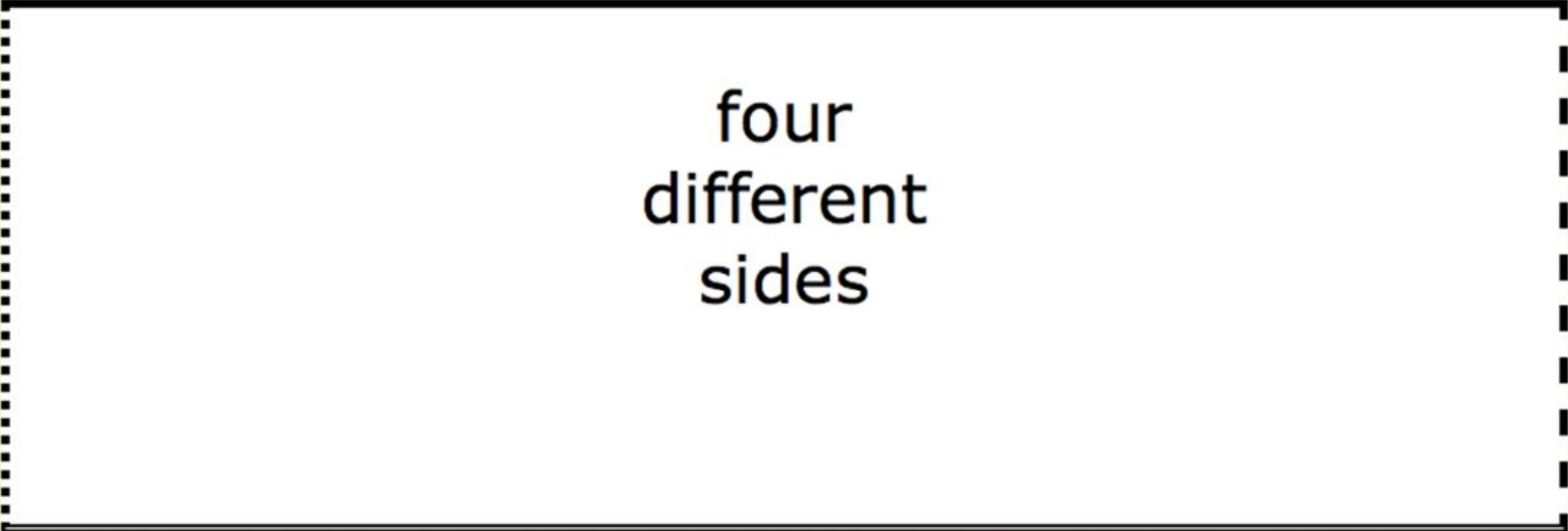
If there is no color specified, the border uses the foreground color of the element (same as the text)

Borders Example

```
div#silly {  
  border-top-style: solid;  
  border-right-style: dashed;  
  border-bottom-style: double;  
  border-left-style: dotted;  
  width: 300px;  
  height: 100px;  
}
```

```
div#silly {  
  border-style: solid dashed double dotted;  
}
```


Borders Example



four
different
sides

Border Width (Thickness)

border-width

Values: `length` | `thin` | `medium` | `thick`

Default: `medium`

Border Width (Thickness)

`border-top-width, border-right-width, border-bottom-width,
border-left-width`

Values: `length | thin | medium | thick`

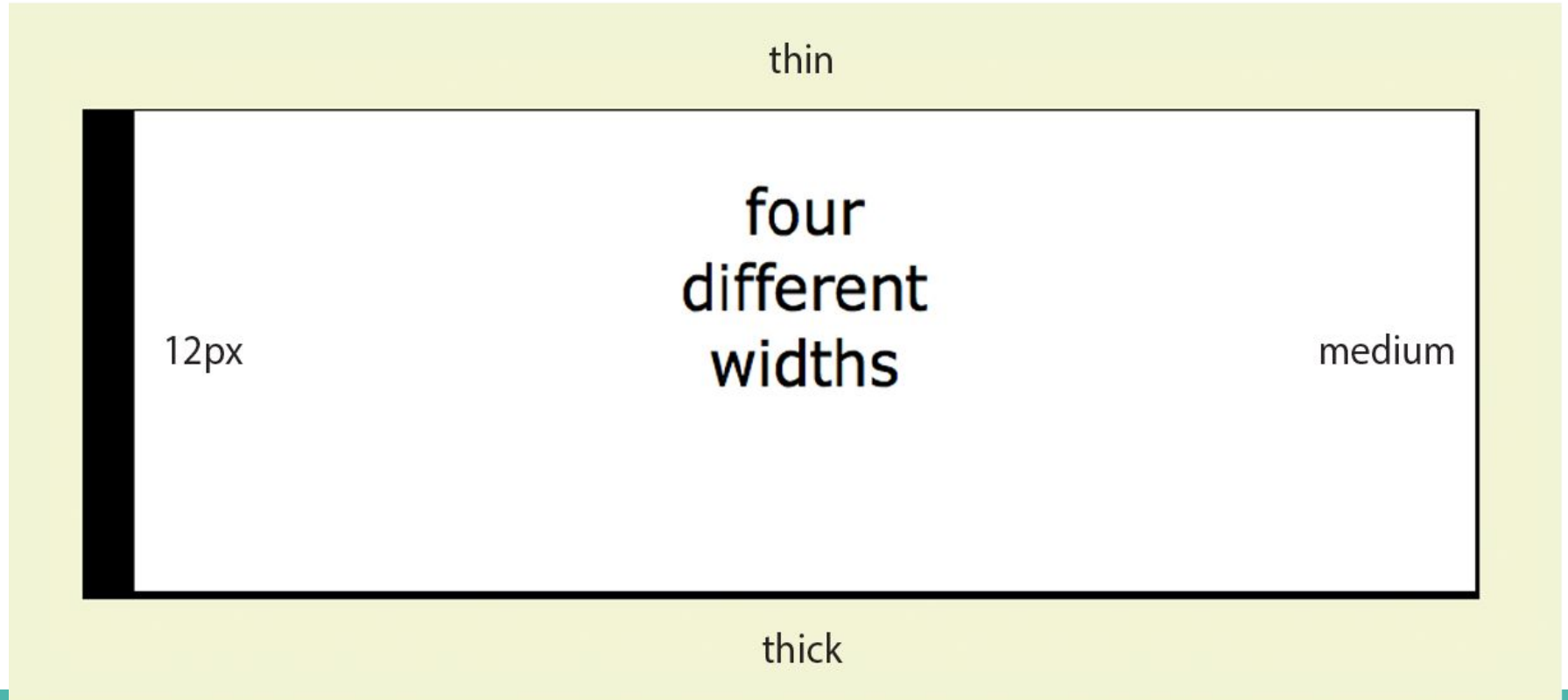
Default: `medium`

Border Width Example

```
div#silly {  
  border-top-width: thin;  
  border-right-width: medium;  
  border-bottom-width: thick;  
  border-left-width: 12px;  
  border-style: solid;  
  width: 300px;  
  height: 100px;  
}
```

```
div#silly {  
  border-width: thin medium thick 12px;  
  border-style: solid;  
}
```

Border Width Example



Border Color

border-color

Values: color name or RGB/HSL value | transparent

Default: the value of the color property for the element

Border Color

`border-top-color, border-right-color, border-bottom-color, border-left-color`

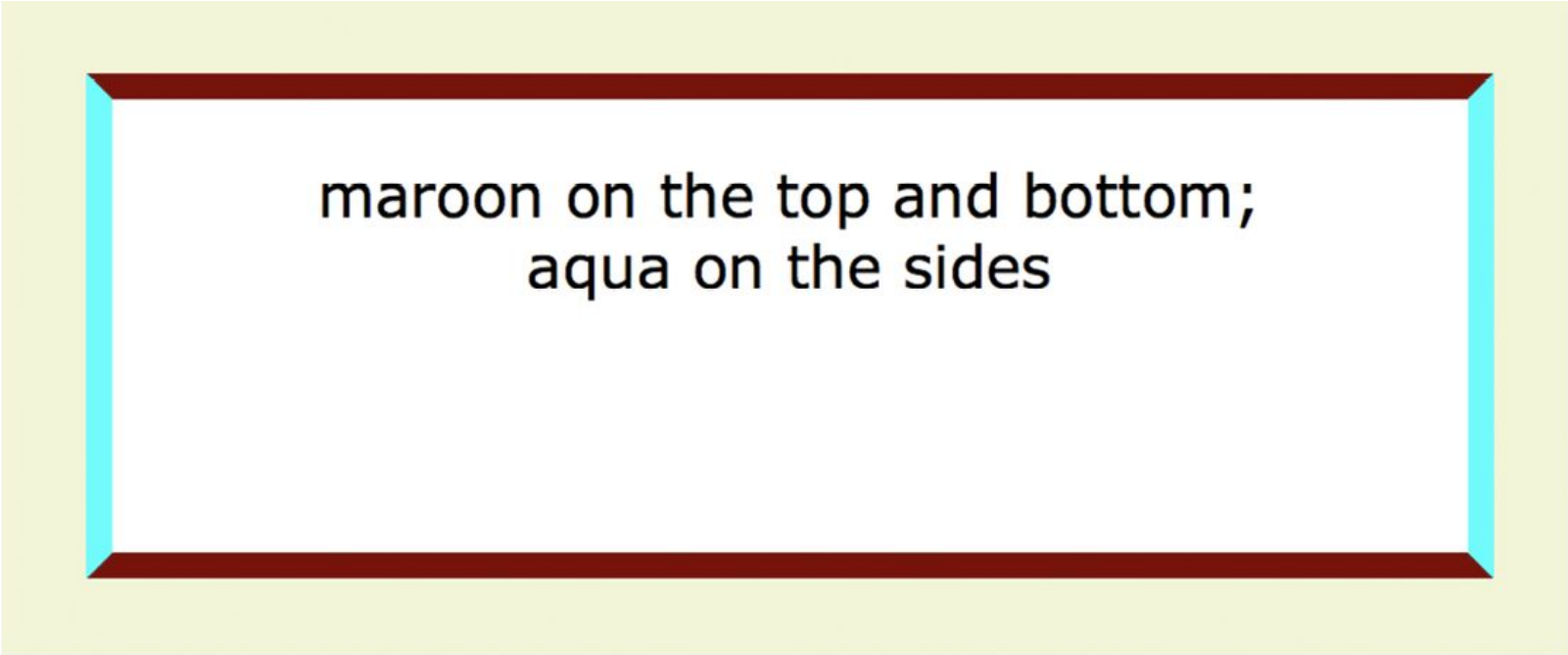
Values: `color name or RGB/HSL value | transparent`

Default: `the value of the color property for the element`

Border Color

```
div#special {  
  border-color: maroon aqua;  
  border-style: solid;  
  border-width: 6px;  
  width: 300px;  
  height: 100px;  
}
```


Border Color



maroon on the top and bottom;
aqua on the sides

Combining Border Style, Width, and Color

border-top, border-right, border-bottom, border-left

Values: `border-style border-width border-color`

Default: defaults for each property

border

Values: `border-style border-width border-color`

Default: defaults for each property

Combining Border Style, Width, and Color

The values for border and the side-specific border properties may include **style**, **width**, and **color** values in any order

```
h1 { border-left: red .5em solid; } /* left border only */  
h2 { border-bottom: 1px solid; } /* bottom border only */  
p.example { border: 2px dotted #663; } /* all four sides */
```

Rounded Corners with border-radius

border-radius

Values: 1, 2, 3, or 4 length or percentage values

Default: 0

Rounded Corners with border-radius

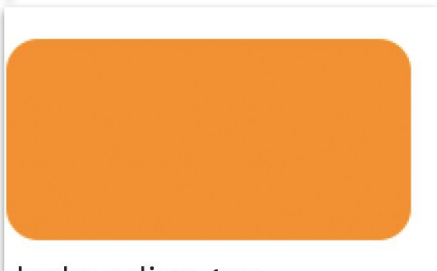
`border-top-left-radius, border-top-right-radius,
border-bottom-right-radius, border-bottom-left-radius`

Values: 1, 2, 3, or 4 length or percentage values

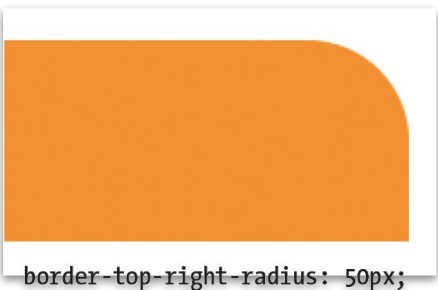
Default: 0

Rounded Corners with border-radius

```
p {  
  width: 200px;  
  height: 100px;  
  background: darkorange;  
}
```



`border-radius: 1em;`



`border-top-right-radius: 50px;`



`border-radius: 50px;`



`border-top-left-radius: 1em;
border-top-right-radius: 2em;
border-bottom-right-radius: 1em;
border-bottom-left: 2em;`

or

`border-radius: 1em 2em;`

Elliptical corners

You can make a corner **elliptical** by specifying two values:

the first for the **horizontal radius** and

the second for the **vertical radius**

A

E



```
border-top-right-radius: 100px 50px;
```

B



```
border-top-right-radius: 50px 20px;  
border-top-left-radius: 50px 20px;
```

C



```
border-radius: 60px / 40px;
```

D



```
border-radius:  
36px 40px 60px 20px/12px 10px 30px 36px;
```


Exercise

Add `tan double border` around the main element

```
padding: 1em;  
border: double 4px #eaddc4;
```

Exercise

Add **rounded corners** to the **main** and **aside** elements

```
border-radius: 25px;
```

Exercise

Adds a **1-pixel solid border** on the **top** of the headline

Add another border that adds a **3-pixel solid border** on the **left**

```
border-top: 1px solid;  
border-left: 3px solid;
```

Exercise

Remove the **underline** for all links

Add a **1-pixel dotted border** to the **bottom** edge of links

```
text-decoration: none;  
border-bottom: 1px dotted;
```

MUFFINS



Every day, we offer a large selection of muffins, including blueberry, multi-berry, bran, corn, lemon-poppyseed, and chocolate. Our muffins are made from scratch each day. Stop by to see our seasonal muffin flavors!

[LEARN MORE ABOUT HOW WE MAKE OUR MUFFINS...](#)

Hours

Margins

A margin is an optional amount of space that you can add on the outside of the border

Margins keep elements from bumping into one another or the edge of the browser window or viewport

The shorthand **margin** property works the same as the **padding** shorthand

Margins

margin

Values: `length` | `percentage` | `auto`

Default: `auto`

Margins

margin-top, margin-right, margin-bottom, margin-left

Values: `length` | `percentage` | `auto`

Default: `auto`

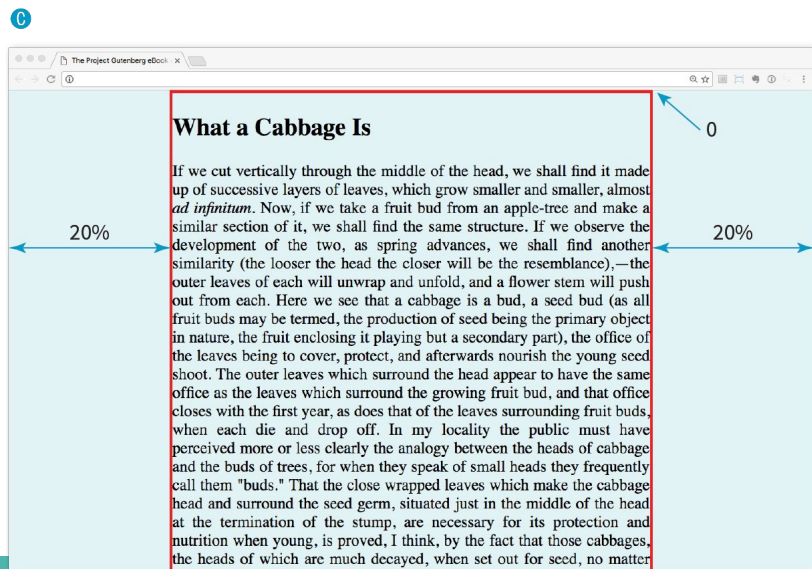
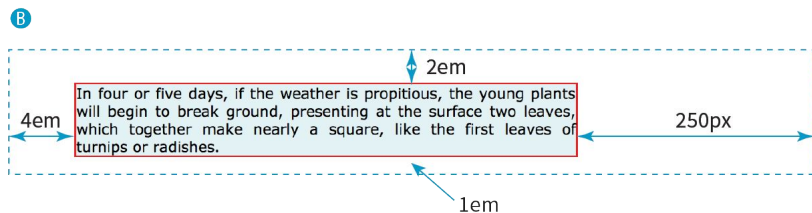
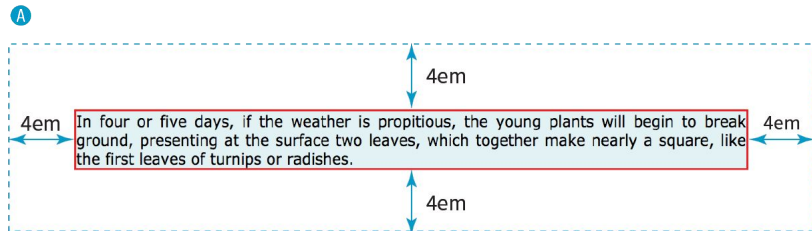
Margins

```
p#A {  
  margin: 4em;  
  border: 2px solid red;  
  background: #e2f3f5;  
}
```

```
body {  
  margin: 0 20%;  
  border: 3px solid red;  
  background-color: #e2f3f5;  
}
```

```
p#B {  
  margin-top: 2em;  
  margin-right: 250px;  
  margin-bottom: 1em;  
  margin-left: 4em;  
  border: 2px solid red;  
  background: #e2f3f5;  
}
```

Margins



Collapsing margins

The **top** and **bottom margins** of neighboring elements collapse

Only the largest value is used

The only time **top** and **bottom margins** don't collapse is for floated or absolutely positioned elements

Margins on the **left** and **right** sides never collapse

Check margin collaps using Chrome DevTool

Margins on inline elements

You can apply **top** and **bottom margins** to **inline text elements** but it won't add vertical space above and below the element, and the height of the line will not change

When you apply **left** and **right margins** to **inline text elements**, margin space will be held clear before and after the text in the flow of the element, even if that element breaks over several lines

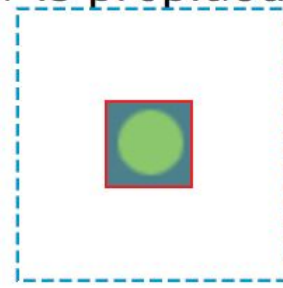
Margins on **replaced inline elements**, such as images, do render on all sides, and therefore do affect the height of the line

```
em { margin: 2em; }
```

In four or five days, if the weather is propitious, the young plants will begin to *break ground*, presenting at the surface two leaves, which together make nearly a square, like the first leaves of turnips or radishes.

```
img { margin: 2em; }
```

In four or five days, if the weather is propitious, the young



plants will begin to break ground, presenting at the surface two leaves, which together make nearly a square, like the first leaves of turnips or radishes.

Negative margins

When you apply a negative margin, the content, padding, and border are moved in the opposite direction that would have resulted from a positive margin value

Negative Margin

```
p.top { margin-bottom: 3em; }
```

Pushes the following paragraph down by 3 ems.

In four or five days, if the weather is propitious, the young plants will begin to break ground, presenting at the surface two leaves, which together make nearly a square, like the first leaves of turnips or radishes.

In four or five days, if the weather is propitious, the young plants will begin to break ground, presenting at the surface two leaves, which together make nearly a square, like the first leaves of turnips or radishes.

```
p.top { margin-bottom: -3em; }
```

The following element moves up by 3 ems.

In four or five days, if the weather is propitious, the young plants will begin to break ground, presenting at the surface two leaves, which together make nearly a square, like the first leaves of turnips or radishes.

In four or five days, if the weather is propitious, the young plants will begin to break ground, presenting at the surface two leaves, which together make nearly a square, like the first leaves of turnips or radishes.

Exercise

Clear out the browser's default margin setting

```
body {  
  margin: 0;  
}  
  
nav ul {  
  margin: 0;  
}
```


Exercise

Add a **margin-top** to **h1**

```
h1 {  
  margin-top: 1.5em;  
}
```

Exercise

Give the **main** element a **2.5% margin** on all sides

For the **aside** element put **1em** on the **top**, **2.5%** on the **right** side, **0** on the **bottom**, and **10% margin** on the **left** side

BLACK GOOSE BAKERY

The delicious baked goods you love at Black Goose Bistro...now available to go!

Fresh from the Oven

BREADS



Our breads are made daily from highest-quality whole grain flour, water, salt, and sourdough starter. Simply and naturally, and never any preservatives. Patience is key to achieving the proper level of fermentation and baking each loaf to perfection. Available in whole grain, sourdough, olive loaf, classic rye, and onion.

[LEARN MORE ABOUT OUR BAKING PROCESS...](#)

MUFFINS



Every day, we offer a large selection of muffins, including blueberry, multi-berry, bran, poppyseed, and chocolate. Our muffins are made from scratch each day. Stop by to see our seasonal muffin flavors.

[LEARN MORE ABOUT HOW WE MAKE OUR MUFFINS...](#)

Hours

MONDAY: 5am to 3pm

TUESDAY: 5am to 3pm

WEDNESDAY: 5am to 3pm

THURSDAY: 5am to 3pm

FRIDAY: 5am to 3pm

SATURDAY: 6am to 4pm

SUNDAY: 6am to 4pm

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Assigning Display Types

The display property defines the type of element box an element generates in the layout

Display type assignment is useful for achieving layout effects while keeping the semantics of the HTML source intact

For example, it is common practice to make li elements display as inline elements to turn a list into a horizontal navigation bar. You may also make an otherwise inline a (anchor) element display as a block in order to give it a specific width and height:

Assigning Display Types

For example, it is common practice to make li elements display as inline elements to turn a list into a horizontal navigation bar

You may also make an otherwise inline a (anchor) element display as a block in order to give it a specific width and height:

```
ul.navigation li { display: inline; }  
ul.navigation li a { display: block; }
```

Assigning Display Types

display

Values: inline | block | run-in | flex | grid | flow | flow-root | list-item | table | table-row-group | table-header-group | table-footer-group | table-row | table-cell | table-column-group | table-column | table-caption | ruby | ruby-base | ruby-text | ruby-base-container | ruby-text-container | inline-block | inline-table | inline-flex | inline-grid | contents | none

Default: inline

Box Drop Shadow

box-shadow

Values: `'horizontal offset'` `'vertical offset'` `'blur distance'` `'spread distance'` `color inset` | `none`

Default: `none`

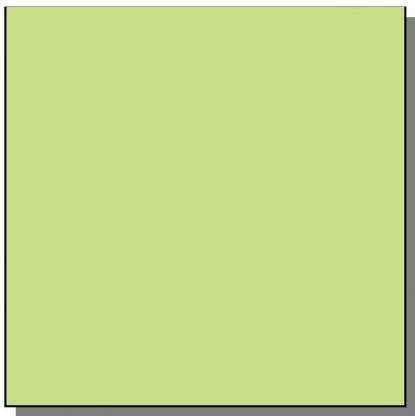
Box Drop Shadow

```
box-shadow: 6px 6px gray; /* A */
```

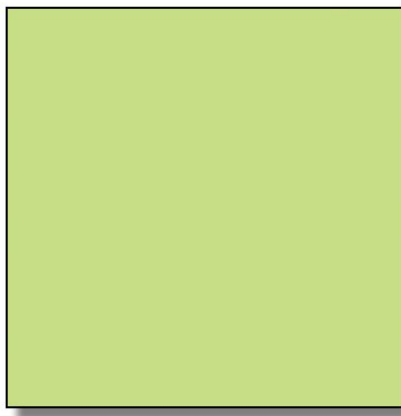
```
box-shadow: 6px 6px 5px gray; /* B */
```

```
box-shadow: 6px 6px 5px 10px gray; /* C */
```

A



B



C

